

PRAKTIKUM SISTEM OPERASI

MODUL 9
SYSCALL XINU



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK
DIREKTORAT KAMPUS PURWOKERTO
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JURNAL/TP

1. Buat syscall baru seperti yang ditunjukkan pada modul syscall poin 9.5! (sertakan Screenshot kode dan hasil run)

a. Membuat syscal baru

```
// Nama: Muhammad Ikhsan Al Hakim
// NIM: 2311104064

#include <xinu.h>

/*-----
 * chname - Dummy syscall
 *-----
 */
syscall chname(pid32 pid, char *newname) {
    intmask mask;

    mask = disable();

    /* Validasi PID*/
    if (isbadpid(pid)) {
        restore(mask);
        return SYSERR;
    }

    /* Validasi nama*/
    if (newname == NULL) {
        restore(mask);
        return SYSERR;
    }

    /* Ubah nama proses*/
    strncpy(proctab[pid].pname, newname, PNMLEN-1);
    proctab[pid].pname[PNMLEN-1] = '\0';

    restore(mask);
    return OK;
}
```

b. Menambahkan prototypes.h

```
extern status ascdate(uint32, char *);

/* in file bufinit.c */
extern status bufinit(void);

/* in file chprio.c */
extern syscall chprio(pid32, pri16);

/* in file chname.c */
extern syscall chname(pid32, char *);

/* in file clkupdate.S */
extern uint32 clkcount(void);

/* in file clkhandler.c */
extern interrupt clkhandler(void);

/* in file clkinit.c */
extern void clkinit(void);

/* in file clkdirp.S */
extern void clkdirp(void);

/* in file close.c */
extern syscall close(did32);

/* in file control.c */
extern syscall control(did32, int32, int32, int32);

/* in file create.c */
extern pid32 create(void *, uint32, pri16, char *, uint32, ...);

/* in file ctxsw.S */
extern void ctxsw(void *, void *);
```

c. xsh_uptime

```
/* xsh_uptime.c - xsh_uptime */

#include <xinu.h>
#include <stdio.h>
#include <string.h>

/*-----
 * xsh_uptime - shell to print the time the system has been up
 *-----
 */
// Muhammad Ikhsan Al Hakim
// 2311104064
shellcmd xsh_uptime(int nargs, char *args[])
{
    uint32  days, hrs, mins, secs; /* Days, hours, minutes, and */
                                     /* seconds since system boot */
    uint32  secperday = 86400;      /* Seconds in a day */
    uint32  secperhr  = 3600;       /* Seconds in an hour */
    uint32  secpermin  = 60;        /* Seconds in a minute */

    /* Nama proses baru */
    char prosesBaru[] = "ProcBaru";

    /* For argument '--help', emit help about the 'uptime' command */
    if (nargs == 2 && strcmp(args[1], "--help", 7) == 0) {
        printf("Use: %s\n\n", args[0]);
        printf("Description:\n");
        printf("\tDisplays time since the system booted\n");
        printf("Options:\n");
        printf("\t--help\t display this help and exit\n");
        return 0;
    }

    /* Subtract number of hours */
    hrs = secs/secperhr;
    secs -= hrs*secperhr;

    /* Subtract number of minutes */
    mins = secs/secpermin;
    secs -= mins*secpermin;

    printf("Xinu has been up ");
    if (days > 0) {
        printf(" %d day(s) ", days);
    }

    if (hrs > 0) {
        printf(" %d hour(s) ", hrs);
    }

    if (mins > 0) {
        printf(" %d minute(s) ", mins);
    }

    if (secs > 0) {
        printf(" %d second(s) ", secs);
    }
    printf("\n");

    /* Syscall baru*/
    kprintf("Mengubah nama proses ID 5 Menjadi %s...\n", prosesBaru);
    chname(5, prosesBaru);

    return 0;
}
```

d. Output

```
Activities Terminal Fri 1 May, 21:03
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help

116598 bytes of Xinu code.
[0x01000000 to 0x0011C775]
148392 bytes of data.
[0x0011FF20 to 0x001442C7]

Obtained IP address 10.0.3.15 (0x0a00030f)

-----MUHAMMAD IKHSAN AL HAKIM-----
XINU
-----2311104064-----

Welcome to Xinu!

xsh $ uptime
Xinu has been up 1 second(s)
Mengubah nama proses ID 5 Menjadi ProcBaru...
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```

2. Perbaiki syscall chprio (xinu/system/chprio.c) dengan memperhatikan validasi input

- Pastikan id adalah angka dari 0 – NPROC (ukuran maks banyaknya proses)
- Pastikan prioritas adalah bilangan yang positif

Compile dan jalankan Xinu dengan syscall yang telah diperbaiki

- make
- clean

```
Activities Terminal Fri 1 May, 21:25
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help

Welcome to Xinu!

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFEF0 8192 1 3
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process recv   20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv   50   4 0x005F7FFC 0x005F79AC 8192 0 0
6 ps         curr   20   5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $ uptime
Xinu has been up 47 second(s)
Mengubah nama proses ID 5 Menjadi ProcBaru...
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFEF0 8192 1 3
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process recv   20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 ProcBaru    recv   50   4 0x005F7FFC 0x005F79AC 8192 0 7
8 ps         curr   20   5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0
```

3. Lakukan hal-hal berikut ini

Edit xsh_uptime.c

- Tambahkan kode berikut

```
secs -= days*secperday;

/* Subtract number of hours */
hrs   = secs/secperhr;
secs -= hrs*secperhr;

/* Subtract number of minutes */
mins  = secs/secpermin;
secs -= mins*secpermin;

printf("Xinu has been up ");
if (days > 0) {
    printf(" %d day(s) ", days);
}

if (hrs > 0) {
    printf(" %d hour(s) ", hrs);
}

if (mins > 0) {
    printf(" %d minute(s) ", mins);
}

if (secs > 0) {
    printf(" %d second(s) ", secs);
}
printf("\n");

/* Syscall baru*/
kprintf("Mengubah prioritas proses ID 5 menjadi 33...\n");
chprio(5, 33);

return 0;
}
```

Compile source code tersebut dengan perintah

- make clean
- make

Jalankan perintah ps

- xsh \$ ps
- perhatikan prioritas proses dengan id = 5

Jalankan uptime

- xsh \$ uptime
- Perhatikan hasil perintah tersebut

Jalankan ps

- xsh \$ ps
- perhatikan prioritas proses dengan id = 5 seharusnya sudah berubah

```

Welcome to Xinu!

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size Msg  Content
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFE00 8192 1 3
1 ipout           wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin           wait  500  0 0x005F9FFC 0x005F9E00 8192 0 0
4 Main process    recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell           recv  50   4 0x005F7FFC 0x005F79AC 8192 0 0
6 ps              curr  20   5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $ uptime
Xinu has been up  8 second(s)
Mengubah prioritas proses ID 5 menjadi 33...

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size Msg  Content
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFE00 8192 1 3
1 ipout           wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin           wait  500  0 0x005F9FFC 0x005F9E00 8192 0 0
4 Main process    recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell           recv  33   4 0x005F7FFC 0x005F79AC 8192 0 7
8 ps              curr  20   5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $

```

Testing chprio syscall yang telah diubah

- Testing prioritas tidak boleh < 0 : Ubah “chprio(5,33)” menjadi “chprio(5,-3)” pada xsh_uptime.c

```

/* Subtract number of hours */
hrs = secs/secperhr;
secs -= hrs*secperhr;

/* Subtract number of minutes */
mins = secs/secpermin;
secs -= mins*secpermin;

printf("Xinu has been up ");
if (days > 0) {
    printf(" %d day(s) ", days);
}

if (hrs > 0) {
    printf(" %d hour(s) ", hrs);
}

if (mins > 0) {
    printf(" %d minute(s) ", mins);
}

if (secs > 0) {
    printf(" %d second(s) ", secs);
}
printf("\n");

/* Syscall baru */
kprintf("Mengubah prioritas proses ID 5 menjadi 33...\n");
chprio(5, -3);

return 0;
}

```

```

Welcome to Xinu!

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size Msg  Content
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFE00 8192 1 3
1 ipout           wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin           wait  500  0 0x005F9FFC 0x005F9E00 8192 0 0
4 Main process    recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell           recv  50   4 0x005F7FFC 0x005F79AC 8192 0 0
6 ps              curr  20   5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $ uptime
Xinu has been up  5 second(s)
Mengubah prioritas proses ID 5 menjadi 33...

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size Msg  Content
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFE9C 8192 1 3
1 ipout           wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin           wait  500  0 0x005F9FFC 0x005F9E00 8192 0 0
4 Main process    recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell           recv  50   4 0x005F7FFC 0x005F79AC 8192 0 7
8 ps              curr  20   5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $

```

- Testing id adalah valid: Ubah “chprio(5,33)” menjadi “chprio(3000,3)”

```

/* Subtract number of hours */
hrs = secs/secperhr;
secs -= hrs*secperhr;

/* Subtract number of minutes */
mins = secs/secpermin;
secs -= mins*secpermin;

printf("Xinu has been up ");
if (days > 0) {
    printf(" %d day(s) ", days);
}

if (hrs > 0) {
    printf(" %d hour(s) ", hrs);
}

if (mins > 0) {
    printf(" %d minute(s) ", mins);
}

if (secs > 0) {
    printf(" %d second(s) ", secs);
}
printf("\n");

/* Syscall baru */
kprintf("Mengubah prioritas proses ID 5 menjadi 33...\n");
chprio(3000, 3);

return 0;
}

```

```
Activities Terminal Fri 1 May, 21:54
xinu@xinu-develop-end: ~/xinu/compile

Welcome to Xinu!

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size Msg  Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE00 8192 1 3
1 ipout       wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin       wait  500  0 0x005F9FFC 0x005F9E00 8192 0 0
4 Main process  recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv  50   4 0x005F7FFC 0x005F79AC 8192 0 0
6 ps         curr  20   5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $ uptime
Xinu has been up 7 second(s)
Mengubah prioritas proses ID 5 menjadi 33...
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size Msg  Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE00 8192 1 3
1 ipout       wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin       wait  500  0 0x005F9FFC 0x005F9E00 8192 0 0
4 Main process  recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv  50   4 0x005F7FFC 0x005F79AC 8192 0 7
8 ps         curr  20   5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0
```

- Hasil dua testing di atas adalah prioritas tidak berubah karena salah argument (dibuktikan dengan menggunakan perintah ps)